

Math 104 — Introduction to Real Analysis

Spring 2026

Practice Final Exam 3

Name: _____

Student ID: _____

This exam contains 9 problems totaling 70 points. Read each problem carefully and justify your answers unless stated otherwise.

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	10	5	5	5	10	10	10	5	10	70
Score:										

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1 Practice Exam

1.1 Question 1 (10 points) — Compactness and Connectedness

Let $E = \{1/n : n \in \mathbb{N}\} \subset \mathbb{R}$ with the standard metric.

- (a) Is E compact? Is $\overline{E} = E \cup \{0\}$ compact? Justify your answers.
- (b) Is E connected? Justify briefly.

1.2 Question 2 (5 points) — Subsequential Limits

True or False (prove or give a counterexample):

If $\{a_n\}$ is a *bounded* sequence of real numbers and every convergent subsequence of $\{a_n\}$ converges to the same limit L , then $a_n \rightarrow L$.

1.3 Question 3 (5 points) — Special Subsets of $\mathbb{R}$

True or False (justify):

There exists a subset of $\mathbb{R}$ that is closed, bounded, uncountable, and contains no open interval.

1.4 Question 4 (5 points) — Differentiability and Rolle's Theorem

Let f and g be continuous on $[a, b]$ and differentiable on (a, b) , with $f(a) = f(b)$ and $g(a) = g(b)$. Determine whether each statement is true or false, and justify briefly.

- (a) There must exist $c \in (a, b)$ with $f'(c) = 0$.
- (b) f' must be continuous on (a, b) .
- (c) There must exist $c \in (a, b)$ with $f'(c) = g'(c)$.

1.5 Question 5 (10 points) — Pointwise vs. Uniform Convergence

Define $f_n(x) = nx^n(1-x)$ on $[0, 1]$.

- (a) Find the pointwise limit $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for each $x \in [0, 1]$.
- (b) Does $f_n \rightarrow f$ uniformly on $[0, 1]$? Justify.
- (c) Compute $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$. Does it equal $\int_0^1 f(x) dx$?

1.6 Question 6 (10 points) — Equicontinuity and Arzelà–Ascoli

Let $\{f_n\}$ be a sequence of differentiable functions on $[0, 1]$ satisfying $|f'_n(x)| \leq 3$ for all n and all $x \in [0, 1]$, and $f_n(0) = 0$ for all n .

- (a) Show that $\{f_n\}$ is equicontinuous on $[0, 1]$.
- (b) Does $\{f_n\}$ necessarily have a uniformly convergent subsequence? Justify.

1.7 Question 7 (10 points) — Riemann Integrability (Thomae's Function)

Define f on $[0, 1]$ by:

$$f(x) = \begin{cases} 1/q & \text{if } x = p/q \text{ in lowest terms, } q \geq 1, \\ 0 & \text{if } x \text{ is irrational,} \end{cases}$$

with $f(0) = 1$.

(a) Show that f is continuous at every irrational point in $(0, 1)$.

(b) Is f Riemann integrable on $[0, 1]$? If so, compute $\int_0^1 f(x) dx$.

1.8 Question 8 (5 points) — Riemann Sums

Compute the following limit by recognizing it as a Riemann sum:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k}.$$

1.9 Question 9 (10 points) — Power Series

Consider the power series $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}$.

(a) Find the radius of convergence R .

(b) Does the series converge at $x = 1$? At $x = -1$? Justify.

(c) Is f continuous on $[-1, 1]$? Justify.

2 Answer Key

2.1 Answer 1 — Compactness and Connectedness

(a) $E = \{1/n : n \in \mathbb{N}\}$ is **not compact**.

The point 0 is a limit point of E (since $1/n \rightarrow 0$), but $0 \notin E$. Therefore E is not closed. In $\mathbb{R}^n$, compact implies closed, so E is not compact.

Alternatively, the open cover $\{B(1/n, \varepsilon_n)\}$ with sufficiently small ε_n has no finite subcover, since E is infinite and discrete (each point $1/n$ is isolated in E).

$\overline{E} = E \cup \{0\}$ is **compact**. It is closed (by definition of closure) and bounded (contained in $[0, 1]$). By the Heine–Borel theorem, closed and bounded subsets of $\mathbb{R}$ are compact.

(b) E is **not connected**.

Let $A = E \cap (-\infty, 3/4)$ and $B = E \cap (3/4, \infty)$. Then $A = \{1/n : n \geq 2\}$ and $B = \{1\}$. Both are nonempty and $E = A \cup B$. Since $\overline{A} \cap B = [0, 3/4] \cap \{1\} = \emptyset$ and $A \cap \overline{B} = A \cap \{1\} = \emptyset$, the sets A and B are separated. Hence E is not connected.

More generally, E is not an interval (e.g., $1/2, 1 \in E$ but $3/4 \notin E$), and connected subsets of $\mathbb{R}$ are precisely the intervals.

2.2 Answer 2 — Subsequential Limits

True.

Since $\{a_n\}$ is bounded, the Bolzano–Weierstrass theorem guarantees that the set E of subsequential limits is nonempty. By hypothesis, $E = \{L\}$.

Recall that $\limsup a_n = \sup E$ and $\liminf a_n = \inf E$. Since $E = \{L\}$:

$$\limsup_{n \rightarrow \infty} a_n = L = \liminf_{n \rightarrow \infty} a_n.$$

Therefore $\lim_{n \rightarrow \infty} a_n = L$.

Remark: Boundedness is essential. The sequence $a_n = n$ has no convergent subsequence at all, so the hypothesis “every convergent subsequence converges to L ” is vacuously true, yet a_n does not converge.

2.3 Answer 3 — Special Subsets of $\mathbb{R}$

True. The Cantor ternary set C is such a set.

- **Closed:** $C = \bigcap_{n=0}^{\infty} C_n$ is an intersection of closed sets, hence closed.
- **Bounded:** $C \subseteq [0, 1]$.
- **Uncountable:** Every element of C has a ternary expansion using only the digits 0 and 2. The map sending each such expansion to a binary number (replacing 2's with 1's) gives a surjection $C \rightarrow [0, 1]$, so $|C| \geq |\mathbb{R}|$.
- **Contains no open interval:** At stage n of the construction, C_n consists of 2^n intervals each of length 3^{-n} , so the total length of C_n is $(2/3)^n \rightarrow 0$. Any open interval has positive length, so no open interval fits inside C .

2.4 Answer 4 — Differentiability and Rolle's Theorem

(a) **True.** This is exactly **Rolle's Theorem**: f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$. Therefore there exists $c \in (a, b)$ with $f'(c) = 0$.

(b) **False.** A function can be differentiable everywhere on (a, b) without f' being continuous. For example, on $[0, 1]$, define:

$$f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Then f is differentiable on $(0, 1)$ (and at 0 by a limit computation), but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ oscillates as $x \rightarrow 0$, so f' is not continuous at 0.

(c) **True.** Define $h = f - g$. Then h is continuous on $[a, b]$, differentiable on (a, b) , and

$$h(a) = f(a) - g(a), \quad h(b) = f(b) - g(b).$$

Since $f(a) = f(b)$ and $g(a) = g(b)$, we get $h(a) = h(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ with $h'(c) = 0$, i.e., $f'(c) - g'(c) = 0$, so $f'(c) = g'(c)$.

2.5 Answer 5 — Pointwise vs. Uniform Convergence

(a) For $x \in [0, 1)$: $x^n \rightarrow 0$, so $f_n(x) = nx^n(1-x) \rightarrow 0$ since $nx^n \rightarrow 0$ for any fixed $|x| < 1$ (exponential decay beats linear growth).

At $x = 1$: $f_n(1) = n \cdot 1^n \cdot 0 = 0$ for all n .

Therefore $f(x) = 0$ for all $x \in [0, 1]$.

(b) The convergence is **not uniform**.

To find $\|f_n\|_\infty = \sup_{x \in [0, 1]} |f_n(x)|$, we maximize $f_n(x) = nx^n(1-x)$. Setting $f'_n(x) = 0$:

$$f'_n(x) = n[nx^{n-1}(1-x) - x^n] = nx^{n-1}[n(1-x) - x] = nx^{n-1}[n - (n+1)x].$$

This is zero at $x^* = \frac{n}{n+1}$. The maximum value is:

$$f_n(x^*) = n \left(\frac{n}{n+1} \right)^n \cdot \frac{1}{n+1} = \frac{n}{n+1} \cdot \left(\frac{n}{n+1} \right)^n \rightarrow 1 \cdot \frac{1}{e} = \frac{1}{e} \neq 0.$$

Since $\|f_n - f\|_\infty \rightarrow 1/e \neq 0$, the convergence is not uniform.

(c) We compute directly:

$$\int_0^1 f_n(x) dx = n \int_0^1 x^n(1-x) dx = n \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2} \right]_0^1 = n \left(\frac{1}{n+1} - \frac{1}{n+2} \right) = \frac{n}{(n+1)(n+2)}.$$

Therefore:

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \lim_{n \rightarrow \infty} \frac{n}{(n+1)(n+2)} = 0.$$

$$\text{And } \int_0^1 f(x) dx = \int_0^1 0 dx = 0.$$

So $\lim_{n \rightarrow \infty} \int_0^1 f_n = \int_0^1 f$, even though convergence is not uniform. This shows that uniform convergence is *sufficient* for interchanging limits and integrals, but not *necessary*.

2.6 Answer 6 — Equicontinuity and Arzelà–Ascoli

(a) Given any $\varepsilon > 0$, set $\delta = \varepsilon/3$. For any n and any $x, y \in [0, 1]$ with $|x - y| < \delta$, the Mean Value Theorem gives some c between x and y with:

$$|f_n(x) - f_n(y)| = |f'_n(c)| |x - y| \leq 3 |x - y| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon.$$

Since $\delta = \varepsilon/3$ works for *all* n and *all* x, y , the family $\{f_n\}$ is equicontinuous.

(b) **Yes.** We verify the hypotheses of the **Arzelà–Ascoli theorem**:

- *Domain is compact:* $[0, 1]$ is compact.
- *Equicontinuous:* Shown in part (a).
- *Pointwise bounded:* For any $x \in [0, 1]$,

$$|f_n(x)| = |f_n(x) - f_n(0)| \leq 3 |x - 0| \leq 3.$$

So $|f_n(x)| \leq 3$ for all n and all x .

By Arzelà–Ascoli, $\{f_n\}$ has a uniformly convergent subsequence.

2.7 Answer 7 — Riemann Integrability (Thomae’s Function)

(a) Let $\alpha \in (0, 1)$ be irrational. We show f is continuous at α .

Given $\varepsilon > 0$, choose an integer N with $1/N < \varepsilon$. Consider all rationals p/q in $[0, 1]$ with $q \leq N$. There are only *finitely many* such rationals (at most N^2 of them). Since α is irrational, α is not equal to any of them. Therefore we can choose $\delta > 0$ such that the interval $(\alpha - \delta, \alpha + \delta)$ contains none of these rationals.

Now if $|x - \alpha| < \delta$:

- If x is irrational, then $|f(x) - f(\alpha)| = |0 - 0| = 0 < \varepsilon$.
- If $x = p/q$ in lowest terms, then $q > N$ (since all rationals with $q \leq N$ are excluded), so $|f(x) - f(\alpha)| = 1/q < 1/N < \varepsilon$.

In both cases $|f(x) - f(\alpha)| < \varepsilon$, so f is continuous at α .

(b) f is **Riemann integrable** on $[0, 1]$, and $\int_0^1 f(x) dx = 0$.

f is discontinuous precisely at the rational points in $[0, 1]$. The set $\mathbb{Q} \cap [0, 1]$ is countable, hence a set of measure zero (given $\varepsilon > 0$, cover the k -th rational with an interval of length $\varepsilon/2^k$; total length $\leq \varepsilon$).

Since f is bounded (by 1) and its set of discontinuities has measure zero, $f \in \mathcal{R}[0, 1]$.

For the value: for any partition P , each subinterval contains irrational points where $f = 0$, so $m_i = \inf f = 0$ on every subinterval. Therefore $L(P, f) = 0$ for every partition, giving $\int_0^1 f = 0$. Since $f \in \mathcal{R}$, the integral equals 0.

2.8 Answer 8 — Riemann Sums

Rewrite the sum by factoring out $1/n$:

$$\sum_{k=1}^n \frac{1}{n+k} = \sum_{k=1}^n \frac{1}{n} \cdot \frac{1}{1+k/n}.$$

This is a Riemann sum for $\int_0^1 \frac{1}{1+x} dx$ with the partition $\{0, 1/n, 2/n, \dots, 1\}$ and right endpoints $x_k = k/n$:

$$\sum_{k=1}^n \frac{1}{n} \cdot \frac{1}{1+k/n} \rightarrow \int_0^1 \frac{dx}{1+x} = [\ln(1+x)]_0^1 = \ln 2 - \ln 1 = \boxed{\ln 2}.$$

2.9 Answer 9 — Power Series

(a) Using the root test, $\limsup_{n \rightarrow \infty} |c_n|^{1/n} = \limsup_{n \rightarrow \infty} (1/n^2)^{1/n}$. Since $n^{1/n} \rightarrow 1$, we have $(1/n^2)^{1/n} = (n^{1/n})^{-2} \rightarrow 1$. Therefore:

$$R = \frac{1}{\limsup |c_n|^{1/n}} = \frac{1}{1} = 1.$$

(b) At $x = 1$: the series becomes $\sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges (it is a p -series with $p = 2 > 1$).

At $x = -1$: the series becomes $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$, which converges absolutely since $\sum \frac{|(-1)^n|}{n^2} = \sum \frac{1}{n^2} < \infty$.

(c) **Yes**, f is continuous on $[-1, 1]$.

For all $x \in [-1, 1]$ and all n :

$$\left| \frac{x^n}{n^2} \right| \leq \frac{1}{n^2} M_n.$$

Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$, the **Weierstrass M -test** implies $\sum x^n/n^2$ converges *uniformly* on $[-1, 1]$.

Each partial sum $S_N(x) = \sum_{n=1}^N x^n/n^2$ is a polynomial, hence continuous. Since the uniform limit of continuous functions is continuous (Theorem 7.12), f is continuous on $[-1, 1]$.